MILITARY SPECIFICATION

GUIDED WEAPON CONTROL
GROUP OK-293A/AWW

This amendment forms a part of Military Specification MIL-G-85231(AS) dated 30 October 1984, and is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

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Delete "3.4.a" in its entirety and substitute:

a. Under standard conditions (see 3.3.10), the control group shall provide the regulated outputs as specified in Table II and the Appendix.

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Add "Appendix":

APPENDIX
DATA WORD REQUIREMENTS FOR
GUIDED WEAPON CONTROL GROUP OK-293A/AWW

10. SCOPE.

10.1 Scope. This appendix details the data word requirements of the Guided Weapon Control Group OK-293A/AWW. This appendix is a mandatory part of this specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS.

20.1 This section is not applicable to this appendix.

30. REQUIREMENTS

30.1 16-BIT digitally encoded word. The control group shall use a 16-BIT digitally encoded word format as shown below:

SYNC	SYNC	ANT SEL	LSB	2ND	MSB	8 SPARES	LAST TWO
1	2	3	4	5	6	7-14	15-16

FSC 1430

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Where LSB is "Least Significant Bit" and MSB is "Most Significant Bit," the aft antenna shall be selected if the third bit is a "one"; and the forward antenna shall be selected if the third bit is a "zero." Eight spares shall be provided for future growth. The last two bits shall be hardwired to a "zero."

30.2 Data. The data shall consist of a 16-bit parallel word with two bits "Sync" and 12 bits of "Data" with the last two bits tied to "zero." The first half of each bit shall be "Sync" with the second half of each bit data being a "one" or "zero." For "Sync," the second half of each bit is "Sync." The clock rate shall be 50 +4.4, -4.0 Hz.

30.3 Clock rate. The clock rate shall be determined by the 1300 Hz voltage controlled oscillator (VCO). The VCO sine wave output shall be converted to a square wave and then divided by 26 to obtain a 50 percent duty cycle. During standard conditions the clock rate is as follows:

a. $\frac{1300\pm5\text{Hz}}{26}$ or 50±0.2 Hz for a "Sync."

b. $\frac{1202\pm5\text{Hz}}{26}$ or 46.23±0.2 Hz for a "Zero."

c. $\frac{1398\pm5\text{Hz}}{26}$ or 53.77±0.2 Hz for a "One."

During service conditions the clock rate is as follows:

a. $\frac{1300\pm16\text{Hz}}{26}$ or 50±0.62 Hz for a "Sync."

b. $\frac{1202\pm16\text{Hz}}{26}$ or 46.23±0.62 Hz for a "Zero."

c. $\frac{1398\pm16\text{Hz}}{26}$ or 53.77±0.62 Hz for a "One."

30.4 Data definition (standard and service conditions). The data shall be defined for standard and service conditions as follows:

a. Standard conditions

(1) A sync bit shall be defined as 1269 to 1331 Hz.
(2) A data zero shall be defined as 1197 to 1250 Hz.
(3) A data one shall be defined as 1351 to 1403 Hz.

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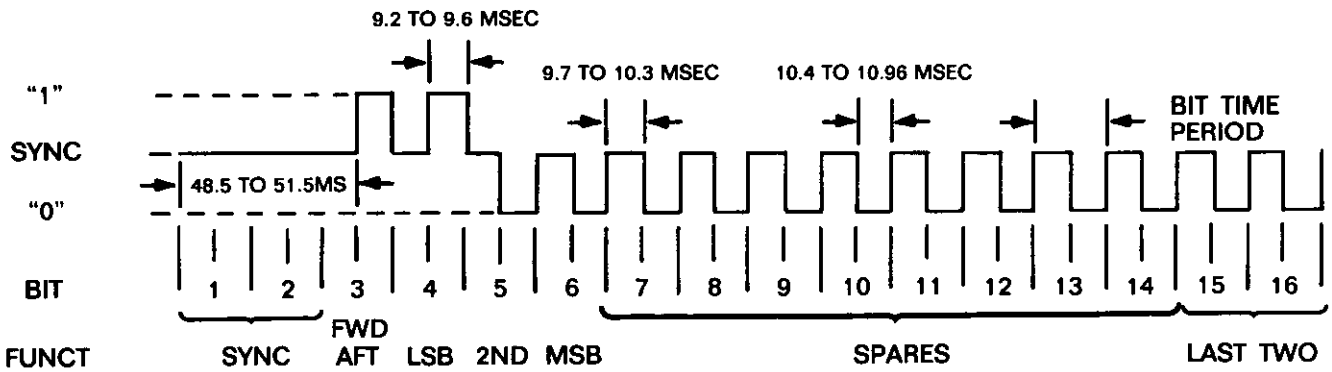
b. Service conditions

(1) A sync bit shall be defined as 1258 to 1342 Hz.
(2) A data zero shall be defined as 1186 to 1250 Hz.
(3) A data one shall be defined as 1351 to 1414 Hz.

30.5 Word format. An example of the word format is specified in figure 7 and as follows:

a. Aft. Ant. shall be a one, the next data bit shall be a one, and all other data bits shall be a zero. The last two data bits, 15 and 16, shall be tied to zero permanently.

30.6 Format logic. Format logic shall be as specified in Table X for presently assigned weapons.



NOTES:

1. The 1 level shall be +4.7 to +5.3 volts dc.
2. The SYNC level shall be +2.35 to +2.65 volts dc.
3. The 0 level shall be 0.0 to +0.30 volts dc.

FIGURE 7. Modulation waveform

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TABLE X. Format logic

Channel Select Switch Letter	FWD AFT	LSB	2ND	MSB	Spares 7-14	Last Two	Channel	Format
A	0 or 1	1	0	0	0	0	A	Phase I
C	0 or 1	0	1	0	0	0	C	Phase I
E	0 or 1	1	1	0	0	0	E	Phase I
F	0 or 1	0	0	1	0	0	A	Phase II
H	0 or 1	1	0	1	0	0	C	Phase II
J	0 or 1	0	1	1	0	0	E	Phase II
K	0 or 1	1	1	1	0	0	None	Future
M	0 or 1	0	0	0	0	0	Manual	For AWW-7 Pod

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